

JUNFEI ZHAN

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Education

University of Pennsylvania

MS in Electrical Engineering

Aug. 2024 – Expected May 2026

Philadelphia, USA

GPA: 4.00/4.00

University of Birmingham & Jinan University

BSc in Applied Mathematics with Information Computing Science

Dual Degree - BSc in Information and Computing Science

Sep. 2020 – June 2024

Birmingham, UK

Guangzhou, CN

GPA: 3.82/4.00 (Birmingham) & 4.01/5.00 (Jinan) **Honors:** First Class (Birmingham)

Relevant Coursework

- Autonomous Driving (A+)
- Graph Neural Networks (A)
- Applied Machine Learning (A)
- Information Theory (A)
- Linear System Theory (A)
- MATLAB Programming (A+)
- Game Theory (A+)
- Applied Statistics (A)
- Convex Optimization (A+)

Honors & Awards

Outstanding Undergraduate Thesis Award, Jinan University, 2024

Outstanding Undergraduate Achievement Award, Jinan University, 2024

First-class Scholarship, Jinan University, 2023 & 2024

China National Scholarship (0.2% - The highest national scholarship for Chinese undergraduates), 2022

Publications

Junfei Zhan, Jiayi Wu, Tengjiao He, Kwan-Wu Chin, "Task Offloading and Approximate Computing in Solar Powered IoT Networks," *IEEE Networking Letters*, vol. 6, no. 1, pp. 26-30, March 2024, doi: 10.1109/LNET.2023.3328893.

Haoxun Shen, **Junfei Zhan**, Tengjiao He, "RL-Enhanced Disturbance-Aware MPC for Fast and Robust UAVs Trajectory Tracking." *Proceedings of the IEEE SMC 2025 Conference* (Accepted, Tsinghua-A).

Junfei Zhan, Weihang Ding, Haoxun Shen, Tengjiao He, "Stochastic Power Modeling and Constrained MDP Optimization for On-Device SLM Inference," *ICASSP 2026 Conference* (Under Review, CCF-B).

Junfei Zhan, Haoxun Shen, Zheng Lin, Tengjiao He, "PRISM: Privacy-Aware Routing for Adaptive Cloud-Edge LLM Inference via Semantic Sketch Collaboration" *AAAI 2026 Conference* (Under Review, CCF-A).

Tengjiao He, **Junfei Zhan**, Kwan-Wu Chin, Fei Song, "Minimizing Maximum Age of Service in Computing Powered Green IoT Networks." *IEEE Transactions on Communications* (Under Review, SCI JCR Q1, CCF-B).

Research Projects

Stochastic Power Modeling and Constrained MDP Optimization for On-Device

SLM Inference

Lead Author, Advisors: Prof. Tengjiao He

Jul. 2025 – Sep. 2025

- Built Hidden Semi-Markov Models (HSMMs) to capture stochastic CPU/GPU power dynamics and phase durations during on-device small language model (SLM) inference.
- Integrated HSMM-based energy estimates with LLM-as-Judge scoring to model the trade-off between inference quality and energy consumption in on-device SLM inference.
- Formulated a constrained Markov decision process (CMDP) with Q-learning to adaptively schedule inference tasks under finite energy budgets and device-level power caps.

Privacy-Aware Routing for Adaptive Cloud-Edge LLM Inference via Semantic Sketch Collaboration

Lead Author, Advisor: Prof. Tengjiao He

Mar. 2025 – Aug. 2025

- Proposed PRISM, a privacy-aware framework that dynamically routes user prompts between cloud, edge, and collaborative modes based on entity-level sensitivity.
- Designed an adaptive two-layer local differential privacy mechanism and a semantic sketch pipeline to balance privacy, utility, and system latency.
- Deployed PRISM on real-world cloud-edge infrastructure, achieving 65% lower energy consumption and $1.54\times$ faster inference over baseline privacy-preserving methods.

Minimizing Maximum Age of Service in Virtualized Green IoT Networks

Research Assistant, Advisor: Prof. Kwan-Wu Chin

May 2024 – Dec. 2024

- Formulated a mixed integer linear programming (MILP) model to optimize the embedding and scheduling of Virtual Network Functions (VNFs) in solar-powered IoT networks, minimizing the maximum age of service (AoS).
- Developed a Receding Horizon Control Optimization (RHCOP) algorithm leveraging Gaussian Mixture Models (GMM) to predict energy arrivals and wireless channel gains for efficient resource allocation.
- Implemented resource-aware embedding of Directed Acyclic Graph (DAG) requests to balance computational loads, manage energy constraints, and enhance the freshness of IoT services.

Competitions

Kaggle: Google - Fast or Slow? Predict AI Model Runtime

Oct. 2023

Prize: Bronze Medal (Top 10%)

- Preprocessed data and built a Graph Convolutional Network (GCN) to predict runtime of graphs and configurations using a machine learning model trained on runtime data.

2022 China Undergraduate Mathematical Contest in Modeling

Sep. 2022

Prize: Provincial First Prize (Top 10%)

- Applied Logistic Regression, Fuzzy C, Principal Component Analysis, and K-Means Clustering to conduct weathering analysis and classification of ancient glass artifacts.

2022 Mathematical Contest in Modeling

Feb. 2022

Prize: Honorable Mention (Top 30%)

- Utilized Empirical Mode Decomposition (EMD) and Long Short-Term Memory (LSTM) neural networks to develop a portfolio strategy for gold and bitcoin.

Internship Experience

JLL (Jones Lang LaSalle)

Aug. 2023 – Oct. 2023

Intern in Data Analytics

Shenzhen, China

- Collected and analyzed data, including tenants, vacancy rates, and business performance metrics, through on-site surveys of 600+ companies and 40+ Grade-A office buildings.
- Utilized SQL for data extraction and Excel for updating databases, ensuring accurate and up-to-date information.
- Developed a program to extract text from large image datasets and efficiently manage text files, significantly improving workflow efficiency.

Extracurricular Activities

Graduate Teaching Assistant

Sep. 2025 – Present

University of Pennsylvania

Philadelphia, PA

- Designed homework assignments for around 125 students in *ESE 5000 Linear System Theory*.
- Held office hours to provide academic support and address individual student needs.
- Led recitations to reinforce lecture material, facilitate problem-solving, and manage Q&A in a large-class setting.

Technical Skills

Programming Languages: C, Python, SQL, MATLAB, R, SPSS

Algorithms: PID Controller, Dijkstra, Dynamic Programming, Receding Horizon Control, Simplex, Big-M Method

Frameworks: PyTorch, Pyomo, scikit-learn